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GRAPH-THEORETIC ANALYSIS OF SCALP EEG FUNCTIONAL CONNECTIVITY FOR EPILEPTIC SEIZURE PREDICTION

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Abstract

Epileptic seizure prediction is a challenging problem in biomedical signal processing, with potential relevance for clinical monitoring and patient safety. This thesis investigates whether pre-ictal changes in brain network topology can be detected from scalp electroencephalography (EEG) recordings using functional connectivity and graph-theoretic analysis.

The proposed pipeline is evaluated on the CHB-MIT scalp EEG dataset. EEG recordings are segmented into baseline and pre-ictal windows preceding seizure onset. For each segment, functional connectivity matrices are computed across multiple frequency bands using coherence, weighted Phase Lag Index, and amplitude envelope correlation. These matrices are interpreted as weighted brain networks from which graph metrics such as clustering coefficient, global efficiency, characteristic path length, small-worldness, modularity, betweenness centrality, and assortativity are extracted.

An initial exploratory analysis on subject chb01 shows that delta-band wPLI mean betweenness centrality decreases consistently before seizures and survives false discovery rate correction in a repeated-measures ANOVA. Since this result is obtained from a single-subject pilot analysis, it is treated as preliminary and used to guide further validation on additional subjects and exploratory prediction experiments.

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Chapter 1

Dataset and Preprocessing

The goal of this stage is to transform raw scalp EEG recordings into a structured set of seizure-centred windows suitable for functional connectivity estimation and graph-theoretic analysis. The pipeline is designed to be reproducible and modular: the dataset is first screened according to explicit eligibility criteria, raw EEG recordings are then preprocessed, standardized temporal windows are extracted around seizure onset, and each window is divided into short epochs. The resulting arrays are used as input for the connectivity computation described in chapter 2.

1.1 CHB-MIT Dataset

The experimental work presented in this thesis is based on the CHB-MIT Scalp EEG Database, a publicly available dataset distributed through PhysioNet and widely used in seizure detection and seizure prediction research [1]. The dataset contains long-term scalp EEG recordings acquired from pediatric patients with intractable epilepsy during clinical monitoring. The recordings provide a valuable resource for studying the temporal evolution of brain activity before and during epileptic seizures.

A note on terminology is necessary because the CHB-MIT dataset distinguishes between subjects and cases. According to the official PhysioNet description, the database consists of 23 cases obtained from 22 pediatric subjects, since case `chb21` corresponds to a later recording session of the same individual as `chb01`. However, the downloadable dataset used in this thesis contains case folders ranging from `chb01` to `chb24`. The additional case `chb24` was added after the original release and is not included in the `SUBJECT-INFO` metadata file. For consistency, the terms *subject*, *case*, and *subject ID* are used throughout the remainder of this thesis to refer to the CHB-MIT dataset identifiers rather than necessarily to unique biological individuals.

The EEG recordings are stored in European Data Format (EDF) files, while seizure onset and offset times are provided through accompanying annotation files and metadata. These annotations enable a seizure-centred analysis, since they allow temporal windows to be defined relative to seizure onset and make it possible to investigate changes in brain dynamics during the preictal period.

The present work adopts a sensor-level approach to functional brain network analysis. Each bipolar EEG channel is treated as a node in a functional network, whereas pairwise functional connectivity estimates define weighted edges between nodes. This approach avoids the need for source reconstruction, which would require subject-specific anatomical information that is not available in the CHB-MIT dataset. At the same time, it imposes an important methodological requirement: network nodes must remain comparable across subjects, seizures, frequency bands, connectivity measures, and temporal windows.

To ensure such comparability, a fixed montage consisting of 22 bipolar EEG channels was used throughout the analysis. Maintaining a consistent set of network nodes is essential for meaningful comparisons of connectivity matrices and graph-theoretical metrics across recordings. Consequently, only recordings containing the complete required montage were considered eligible for further analysis.

1.2 Dataset Screening and Eligibility Criteria

Not all annotated seizures in the original dataset satisfied the requirements of the present study. Therefore, a dedicated screening procedure was performed before preprocessing in order to identify recordings that provided both the necessary temporal context and the complete channel configuration.

A seizure was retained only if it satisfied two eligibility requirements. First, at least $540\,\mathrm{s}$ of EEG data had to be available before seizure onset. This corresponds to the $9\,\mathrm{min}$ preictal interval used to extract three consecutive $3\,\mathrm{min}$ windows before the seizure. Second, the recording had to contain the complete fixed 22-channel bipolar montage used in this thesis. Since each channel is later treated as a graph node, missing channels would produce graphs with different node sets and would make graph metrics incomparable across seizures.

The overall screening procedure is summarized in figure 1.1. The exclusion categories are not mutually exclusive: some seizures failed both the temporal and channel requirements.

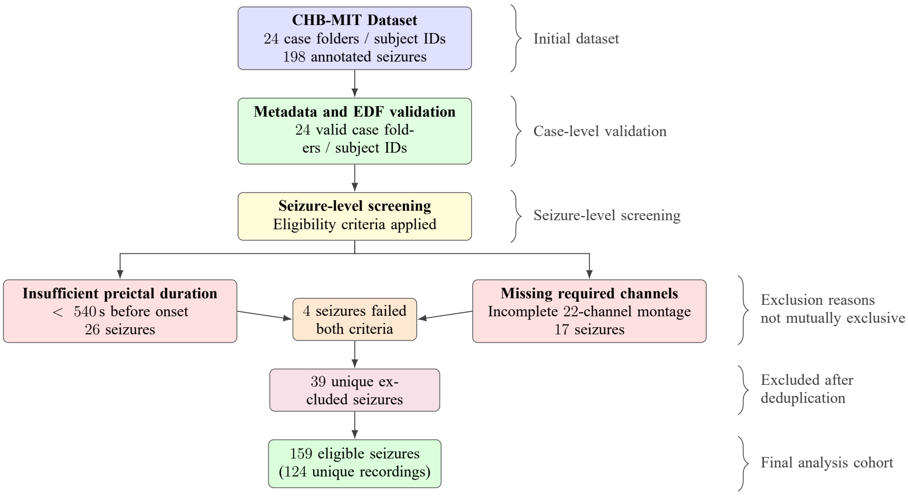

Figure 1.1: Flowchart of the dataset screening procedure and seizure eligibility criteria.

The final screening outcome is reported in table 1.1. In total, 198 annotated seizures were screened. Of these, 159 seizures were retained and 39 were excluded. The retained seizures corresponded to 124 unique EEG recordings, and every screened case folder / subject ID contributed at least one usable seizure.

Screening quantity	Count
Case folders / subject IDs screened	24
Case folders / subject IDs with seizure annotations	24
Case folders / subject IDs with readable metadata	24
Case folders / subject IDs with at least one usable seizure	24
Annotated seizures screened	198
Usable seizures retained	159
Excluded seizures	39
Unique usable recordings	124

Table 1.1: Summary of dataset screening and seizure eligibility.

Among the excluded seizures, 26 had less than 540 s of available preictal data and 17 were missing one or more channels from the required bipolar montage. Since 4 seizures failed both criteria, the sum of the individual exclusion reasons is greater than the number of unique excluded seizures.

The distribution of usable and excluded seizures across case folders is shown in figure 1.2. The figure shows that the dataset is not balanced across case folders: some contain only a few annotated seizures, whereas others contain substantially more. This imbalance is important for the later statistical analysis, because seizures from the same case folder are not fully independent observations.

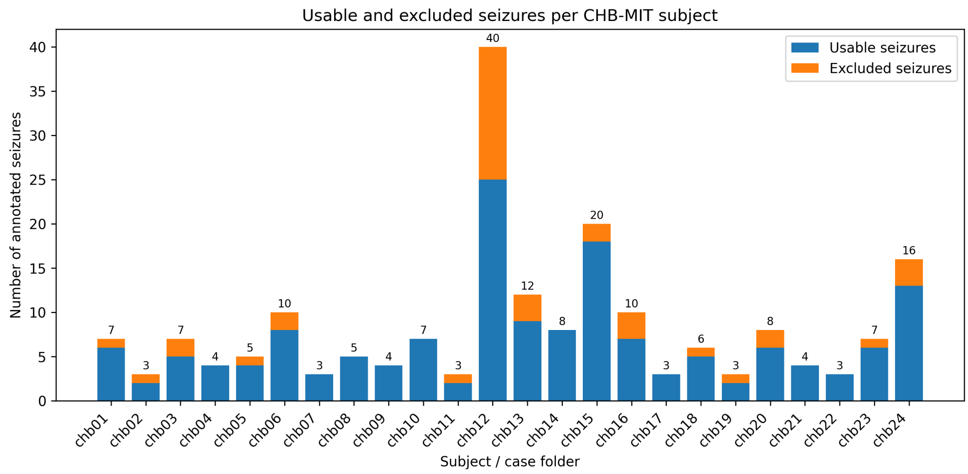

Figure 1.2: Number of usable and excluded annotated seizures for each CHB-MIT case folder / subject ID after dataset screening. The number above each bar indicates the total number of annotated seizures for that case.

1.3 EEG Preprocessing

After dataset screening, all usable seizure-linked recordings were passed to a standardized preprocessing pipeline. The purpose of this stage was to transform the raw EDF recordings into a consistent set of EEG recordings with the same sampling frequency, channel set, channel order, and filtering characteristics. This was necessary because the subsequent connectivity and graph-theoretical analysis requires comparable network nodes across all subjects, seizures, and time windows.

1.3.1 Selection of Recordings for Preprocessing

The preprocessing pipeline used the seizure eligibility table produced during dataset screening as its input. Only recordings associated with at least one usable seizure were selected. If multiple usable seizures occurred in the same EDF recording, the recording was processed only once and reused during the later segmentation stage. This avoided redundant preprocessing while preserving the link between each seizure, subject/case folder, and original recording file.

Each selected EDF file was loaded into memory using MNE-Python. The absolute measurement date stored in the EDF metadata was removed before saving the processed file, because some CHB-MIT headers contain dates that are outside the range accepted by the FIF file format. This operation does not affect the analysis, since all seizure annotations and window definitions in this thesis are expressed in seconds relative to the beginning of each recording rather than in absolute calendar time.

1.3.2 Channel Selection and Montage Standardization

A fixed set of 22 bipolar EEG channels was retained for all recordings. These channels define the graph nodes used in the functional connectivity analysis. The required montage was:

FP1-F7	F7-T7	T7-P7	P7-O1
FP1-F3	F3-C3	C3-P3	P3-O1
FP2-F4	F4-C4	C4-P4	P4-O2
FP2-F8	F8-T8	T8-P8	P8-O2
FZ-CZ	CZ-PZ	P7-T7	T7-FT9
FT9-FT10	FT10-T8		

After loading each recording, channel names were normalized to a canonical format and the recording was checked for the presence of all required channels. Recordings missing one or more channels from the required bipolar montage were excluded during dataset screening. Consequently, all recordings retained for preprocessing contained the complete 22-channel montage.

After validation, the channels were explicitly reordered according to the fixed montage. This ensured that the same channel index corresponded to the same bipolar derivation in every preprocessed recording and in every later connectivity matrix.

1.3.3 Handling of Duplicate `T8-P8` Channels

Some CHB-MIT recordings contain duplicated versions of the `T8-P8` channel. When loaded by MNE-Python, these duplicated channels may appear with suffixes such as `T8-P8-0` and `T8-P8-1`. To preserve a single canonical version of this bipolar channel, the duplicate channel `T8-P8-1` was removed. The retained channel `T8-P8-0` was then renamed to the canonical label `T8-P8`.

This procedure allowed recordings with this naming convention to be retained while still enforcing the same 22-channel montage across the full analysis. Importantly, the duplicate channel was handled before the final channel check and reordering step, so that the output always contained exactly one `T8-P8` channel.

1.3.4 Sampling Frequency Verification

All retained recordings were verified to have a sampling frequency of 256 Hz. No resampling was applied. Recordings with a mismatched sampling frequency would have been rejected during preprocessing, but no such mismatch was observed among the retained usable recordings.

1.3.5 Filtering

The EEG signals were filtered at the recording level before segmentation. A finite impulse response (FIR) band-pass filter was applied between 0.5 Hz and 47 Hz. The lower cutoff was used to attenuate slow drifts and very low-frequency baseline fluctuations, consistent with previous quantitative EEG preprocessing approaches in epilepsy research [2]. The upper cutoff was chosen to match the frequency range required for the present connectivity analysis and is consistent with the 47 Hz upper cutoff used by Vecchio et al. in a pre-seizure graph-theoretical EEG connectivity study [3].

The highest frequency band analyzed later in the pipeline was the beta2 band (20 Hz to 30 Hz). Therefore, the 47 Hz upper cutoff retained the full frequency content required for all analyzed bands while suppressing higher-frequency activity outside the scope of this study. Since the upper cutoff frequency was below the 60 Hz power-line frequency, no additional 60 Hz notch filter was applied in the final preprocessing pipeline.

Filtering was performed on the continuous recording before seizure-centred windows were extracted. This avoids applying filters separately to short segments and reduces the risk of introducing window-specific filtering artifacts.

1.3.6 Artifact Handling

No additional manual or automatic artifact rejection was performed during this preprocessing stage. This choice was made to keep the preprocessing procedure deterministic and reproducible across the full dataset. However, it also means that residual artifacts may still be present in some retained recordings. This limitation is considered when interpreting the connectivity and graph-theoretical results.

1.3.7 Saving Preprocessed Recordings

After channel standardization, sampling-frequency verification, and filtering, each preprocessed recording was saved as an intermediate MNE FIF file and reused during the segmentation stage. Saving these intermediate files allowed preprocessing to be separated from later analysis steps and ensured that all subsequent stages used the same standardized recordings.

A preprocessing log was generated to record the processing status of each selected recording, including the subject/case identifier, recording file, sampling frequency, number of retained channels, and any error messages.

1.4 Temporal Window Definition

After preprocessing, seizure-centred temporal windows were extracted from each eligible seizure recording. The aim of this step was to describe how functional connectivity and graph-theoretical properties evolve during the minutes preceding seizure onset. For this reason, the analysis was organized around one baseline window and three consecutive preictal windows.

The temporal structure was inspired by the design used by Vecchio et al., who investigated graph-theoretical changes during the 9 min period preceding seizure onset by dividing this interval into three consecutive 3 min windows [3]. The present study adopted the same preictal window structure, but applied it to a larger scalp EEG dataset and to multiple connectivity measures and graph metrics.

For each seizure, four windows were defined:

- **Baseline**: a 3 min interictal window selected from the same recording, at least 600 s away from any annotated seizure activity;
- **T0**: the interval from 9 min to 6 min before seizure onset;
- **T1**: the interval from 6 min to 3 min before seizure onset;
- **T2**: the interval from 3 min to 0 min before seizure onset.

The baseline window was used as a reference period representing interictal EEG activity within the same recording. To reduce contamination by seizure-related activity, baseline segments were required to be at least 600 s away from all annotated seizures in that recording. The baseline selection therefore used all known seizure intervals in the recording, including seizures that were not retained for the final analysis, because these intervals still represent ictal periods.

The three preictal windows T0, T1, and T2 covered the final 9 min before seizure onset. This organization allowed the analysis to compare baseline activity with progressively later preictal periods and to examine whether network properties changed as seizure onset approached. The use of fixed-duration windows also ensured that connectivity estimates were computed from equal amounts of data across all seizures and conditions.

Each window had a duration of 180 s. After extraction, each window was divided into non-overlapping 2 s epochs. Since the sampling frequency was 256 Hz, each epoch contained 512 samples. Thus, each saved window had the shape

$$90 \times 22 \times 512,$$

corresponding to 90 epochs, 22 EEG channels, and 512 samples per epoch. These epoch-level representations were then used for the computation of frequency-specific functional connectivity matrices.

1.5 Frequency Band Decomposition

Functional connectivity was computed separately within predefined EEG frequency bands. This was done because epileptic brain dynamics may affect different frequency ranges in different ways, and graph-theoretical properties can depend on the spectral content used to define the functional network.

The frequency bands used in this thesis followed the band structure adopted by Vecchio et al. [3]. In particular, the alpha and beta ranges were subdivided into narrower bands, allowing potential differences between lower and higher alpha or beta activity to be examined separately. The analyzed frequency bands were:

Frequency band	Frequency range
Delta	2 Hz to 4 Hz
Theta	4 Hz to 8 Hz
Alpha1	8 Hz to 10 Hz
Alpha2	10 Hz to 13 Hz
Beta1	13 Hz to 20 Hz
Beta2	20 Hz to 30 Hz

Table 1.2: Frequency bands used for functional connectivity analysis.

Although gamma-band activity is often considered in EEG studies, it was not included in the present analysis. The frequency decomposition was predefined to match the study that motivated the temporal and spectral structure of this work, which focused on delta, theta, alpha, and beta sub-bands. In addition, gamma-band scalp EEG is more vulnerable to contamination by muscle activity and other high-frequency artifacts. Since no dedicated artifact rejection procedure was applied in this pipeline, the analysis was restricted to frequencies up to the beta2 band, ending at 30 Hz.

This restriction also kept the number of frequency bands and statistical tests manageable. Gamma-band connectivity is therefore left as a possible extension for future work rather than being included in the main analysis.

Chapter 2

Connectivity and Graph Features

2.1 Overview of Network Construction

After preprocessing and temporal segmentation, each seizure-related EEG window was represented as a three-dimensional array containing 90 non-overlapping epochs, 22 bipolar EEG channels, and 512 samples per epoch. These segmented EEG windows formed the input to the functional connectivity pipeline. The purpose of this stage was to transform each time-windowed EEG segment into a set of functional brain networks that could be summarized using graph-theoretical measures.

In this thesis, a functional brain network is defined as a weighted graph constructed from pairwise statistical relationships between EEG channels. Each of the 22 bipolar channels corresponds to one graph node. The edges between nodes represent functional connectivity values estimated between pairs of channels. Therefore, for each seizure window, frequency band, and connectivity method, the pipeline produced one square connectivity matrix of size 22 $\times$ 22. Each entry of this matrix represents the estimated functional coupling between two EEG channels.

Connectivity was computed separately for each temporal window, frequency band, and connectivity method. The four temporal windows were

Baseline, T0, T1, and T2, as defined in section 1.4. The six analyzed frequency bands were delta, theta, alpha1, alpha2, beta1, and beta2, as defined in section 1.5. For each saved EEG segment, three connectivity methods were applied: magnitude-squared coherence, weighted phase-lag index, and amplitude envelope correlation. Consequently, each saved window produced 18 functional connectivity matrices.

The complete network construction process can therefore be summarized as follows. First, each segmented EEG window was decomposed according to the predefined frequency bands. Second, functional connectivity was estimated between all pairs of channels using each connectivity method. Third, the resulting connectivity matrices were validated to ensure that they had the expected 22×22 shape, contained only finite values, and were symmetric. Finally, each matrix was interpreted as a weighted undirected network and passed to the graph feature extraction stage.

This organization allowed the analysis to examine whether network properties changed across time windows before seizure onset, whether these changes were frequency-specific, and whether they depended on the connectivity method used to define the network. It also preserved the link between each graph, the corresponding subject/case identifier, seizure, temporal window, frequency band, and connectivity method, which was essential for the later statistical analysis.

2.2 Functional Connectivity Estimation

Functional connectivity was estimated between all pairs of EEG channels for each saved temporal window and frequency band. Three complementary connectivity measures were used: magnitude-squared coherence, weighted phase-lag index (wPLI), and amplitude envelope correlation (AEC). These methods were selected because they capture different aspects of statistical dependence between EEG signals: coherence reflects frequency-specific linear coupling, wPLI emphasizes consistent non-zero phase-lag synchronization, and AEC captures co-fluctuations in band-limited amplitude envelopes [4, 5, 6].

The use of connectivity-derived networks is consistent with previous epilepsy studies that model EEG or intracranial EEG activity as functional graphs, where nodes represent recording sites and edges represent pairwise functional coupling [3, 2].

For each segmented EEG window, connectivity was computed separately for the six predefined frequency bands described in section 1.5. The output of each method was a symmetric 22×22 matrix, where rows and columns corresponded to the fixed bipolar EEG channels. The diagonal entries were set to one before saving, since each channel is perfectly connected with itself by definition; self-connections were later removed before graph-theoretical analysis.

2.2.1 Coherence

Magnitude-squared coherence was used to estimate frequency-specific linear coupling between pairs of EEG channels [7, 3]. For two signals $x_i(t)$ and $x_j(t)$, coherence at frequency f can be expressed as

$$C_{ij}(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f)S_{jj}(f)},$$

where $S_{ij}(f)$ is the cross-spectral density between channels i and j, and $S_{ii}(f)$ and $S_{jj}(f)$ are the corresponding auto-spectral densities. The value of coherence lies between zero and one, with higher values indicating stronger linear coupling at the given frequency.

In the implemented pipeline, cross-spectral and auto-spectral densities were estimated using Welch-based spectral estimation. For each $2\,\text{s}$ epoch, the spectral estimates were computed using a segment length of 256 samples and an overlap of 128 samples. Coherence values were then averaged across all epochs and across all frequency bins belonging to the selected frequency band.

This produced one coherence value for each channel pair, frequency band, and temporal window.

The resulting coherence matrices were clipped to the interval $[0, 1]$ and validated before being saved. Because coherence is a symmetric pairwise measure, the resulting matrices were symmetric by construction and represented weighted undirected functional networks.

2.2.2 Weighted Phase-Lag Index

The weighted phase-lag index was used to estimate phase-based synchronization while reducing the influence of zero-lag coupling [5, 2]. This is important in scalp EEG, where instantaneous coupling can arise from volume conduction, common reference effects, or shared sources. Unlike coherence, which can be influenced by both zero-lag and non-zero-lag relationships, wPLI emphasizes consistent phase differences with a non-zero imaginary component.

Before computing wPLI, the EEG epochs were band-pass filtered within the frequency band of interest using a fourth-order Butterworth filter. The analytic signal was then obtained using the Hilbert transform. For a pair of analytic signals, the cross-spectrum-like quantity was computed and its imaginary part was used to estimate wPLI:

$$\mathrm{wPLI}_{ij} = \frac{\left|\mathbb{E}\left[\mathrm{Im}(Z_{ij})\right]\right|}{\mathbb{E}\left[\left|\mathrm{Im}(Z_{ij})\right|\right]},$$

where Z_{ij} denotes the product of the analytic signal from channel i and the complex conjugate of the analytic signal from channel j, and $\mathrm{Im}(\cdot)$ denotes the imaginary part. The expectation was computed across epochs and time samples.

The resulting wPLI values range from zero to one. Values close to zero indicate weak or inconsistent phase-lagged coupling, whereas values closer to one indicate more consistent non-zero phase-lag synchronization between two channels. As with coherence, the output was a symmetric 22×22 connectivity matrix for each temporal window and frequency band.

2.2.3 Amplitude Envelope Correlation

Amplitude envelope correlation was used to estimate coupling between band-limited amplitude fluctuations. While coherence and wPLI focus on spectral or phase-based relationships, AEC captures whether the amplitude envelopes of two EEG channels fluctuate together over time within a given frequency band [6, 8].

For each frequency band, the EEG epochs were first band-pass filtered using the same fourth-order Butterworth filtering procedure used for wPLI. The analytic signal was then computed with the Hilbert transform, and the amplitude envelope of each channel was obtained by taking the absolute value of the analytic signal. For channel i, the amplitude envelope can be written as

$$A_i(t) = |z_i(t)|,$$

where $z_i(t)$ is the analytic signal of the band-limited EEG signal.

For each epoch, Pearson correlation coefficients were computed between the amplitude envelopes of all channel pairs. The final AEC matrix for a given window and frequency band was obtained by averaging these epoch-level correlation matrices. Unlike coherence and wPLI, AEC values may be positive or negative, because they are based on correlation coefficients. Therefore, the values were clipped to the interval $[-1, 1]$ before validation and saving.

AEC provides a complementary view of functional connectivity because it is sensitive to co-fluctuations in signal amplitude rather than phase consistency alone. However, because no orthogonalization or source reconstruction was applied, AEC may still be influenced by zero-lag correlations and shared sensor-level activity. This limitation is considered when interpreting connectivity differences across temporal windows.

2.3 Adjacency Matrix Construction

After functional connectivity estimation, each temporal window, frequency band, and connectivity method was represented by a square adjacency matrix. These matrices provided the bridge between EEG signal analysis and graph-theoretical feature extraction. In this representation, EEG channels were treated as graph nodes and pairwise connectivity estimates were treated as weighted edges.

For each saved EEG segment, the pipeline produced one adjacency matrix for each combination of frequency band and connectivity method. Since the analysis used 6 frequency bands and 3 connectivity methods, each temporal window produced 18 adjacency matrices. Each matrix had dimensions 22 $\times$ 22, corresponding to the fixed 22-channel bipolar montage described in section 1.3.2.

2.3.1 Weighted Matrices

The connectivity matrices were treated as weighted adjacency matrices. For a given seizure, temporal window, frequency band, and connectivity method, the matrix can be written as

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N} \\ A_{21} & A_{22} & \cdots & A_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ A_{N1} & A_{N2} & \cdots & A_{NN} \end{bmatrix},$$

where $N = 22$ is the number of EEG channels. Each off-diagonal element A_{ij} represents the estimated functional connectivity between channel i and channel j. Larger values indicate stronger estimated coupling between the corresponding pair of channels.

All connectivity matrices were required to be square, finite, and symmetric before being accepted by the pipeline. Symmetry is expected because the connectivity measures used in this thesis are pairwise undirected measures: the coupling between channel i and channel j is the same as the coupling between channel j and channel i. Therefore, each matrix represents an undirected weighted network.

The diagonal entries of the connectivity matrices were set to one during connectivity estimation, since each channel is perfectly connected with itself. However, self-connections are not meaningful for graph-theoretical topology. Consequently, self-loops were removed before graph metrics were computed by setting the diagonal entries to zero in the graph construction stage.

The range of matrix values depended on the connectivity method. Coherence and wPLI produced non-negative values in the interval $[0, 1]$. AEC was based on correlation coefficients and could therefore produce values in the interval $[-1, 1]$. Negative AEC values were retained when computing mean connectivity, but negative values were not used as topology weights during graph-metric calculation, as described in the graph feature extraction section.

2.3.2 Channel-to-Node Mapping

A fixed channel-to-node mapping was used for all adjacency matrices. This was necessary because graph metrics are only comparable across seizures and subjects if the same node index always refers to the same EEG derivation. The CHB-MIT dataset contains recordings with montage variability, and previous CHB-MIT studies have therefore restricted analysis to fixed channel sets in order to reduce heterogeneity across recordings [9]. In the present thesis, a stricter screening strategy was used: recordings were retained only when the complete predefined 22-channel bipolar montage was available. This preserved a larger sensor-level network while ensuring that the node definitions remained identical across subjects, seizures, temporal windows, frequency bands, and connectivity methods.

The channel order was inherited from the standardized preprocessing montage and was kept fixed during segmentation, connectivity estimation, and graph feature extraction. For readability, table 2.1 reports the mapping using one-based node numbering. Internally, the computational implementation used zero-based Python indices, but the ordering was identical.

Node	Channel	Node	Channel
1	`FP1-F7`	12	`P4-O2`
2	`F7-T7`	13	`FP2-F8`
3	`T7-P7`	14	`F8-T8`
4	`P7-O1`	15	`T8-P8`
5	`FP1-F3`	16	`P8-O2`
6	`F3-C3`	17	`FZ-CZ`
7	`C3-P3`	18	`CZ-PZ`
8	`P3-O1`	19	`P7-T7`
9	`FP2-F4`	20	`T7-FT9`
10	`F4-C4`	21	`FT9-FT10`
11	`C4-P4`	22	`FT10-T8`

Table 2.1: Fixed channel-to-node mapping used for all functional connectivity matrices and graph-theoretical analyses.

Because this mapping was fixed, the same matrix position always represented the same pair of bipolar EEG channels. For example, the value at position $(1, 2)$ represented the connection between `FP1-F7` and `F7-T7` in every matrix, regardless of subject, seizure, window, frequency band, or connectivity method.

2.3.3 Frequency- and Window-Specific Networks

The network construction procedure was repeated independently for each temporal window and frequency band. Therefore, the analysis did not produce a single network per seizure, but rather a structured collection of networks describing the same seizure from different temporal, spectral, and methodological perspectives.

For each seizure, the possible temporal windows were Baseline, T0, T1, and T2. For each window, connectivity was computed in the delta, theta, alpha1, alpha2, beta1, and beta2 bands. For each frequency band, three connectivity methods were applied: coherence, wPLI, and AEC. A single adjacency matrix can therefore be identified by the tuple

$$(\text{subject},\ \text{seizure},\ \text{window},\ \text{band},\ \text{method}).$$

This organization preserved the full experimental structure of the analysis. It allowed graph metrics to be compared across preictal windows, across frequency bands, and across connectivity methods while retaining the identity of the subject/case folder and seizure from which each network was derived.

The resulting adjacency matrices were saved and subsequently passed to the graph feature extraction stage. At that stage, the matrices were cleaned, thresholded, converted into graph objects, and summarized using global network-level metrics.

2.4 Graph-Theoretical Metrics

Before computing topology-based graph metrics, each matrix was prepared in a standardized way. First, the matrix was symmetrized and the diagonal was set to zero, removing self-loops. For topology-based metrics, negative connectivity values were also set to zero. This was done because the graph metrics used in this thesis interpret edge weights as non-negative connection strengths. In particular, shortest-path metrics converted connection strength into distance, so negative weights could not be interpreted as valid positive distances in the same way as positive connectivity values [10, 11]. However, negative values were preserved when computing mean connectivity, so that the average connectivity measure still reflected the original signed connectivity values when applicable.

A proportional threshold was then applied to the cleaned matrix. Specifically, the strongest positive edges were retained up to a target density of 20 %. Proportional thresholding was used to impose the same intended graph density across subjects, seizures, temporal windows, frequency bands, and connectivity methods. This reduced the influence of weak connections that may be more susceptible to noise while making graph metrics more comparable across matrices [12, 13, 14].

The 20 % density was selected a priori as a compromise between sparsity and network preservation. A lower threshold would retain fewer edges and could increase the risk of disconnected or unstable graphs, whereas a much higher threshold would retain more weak connections and reduce the effect of network sparsification. Since there is no universally accepted density threshold for EEG functional connectivity graphs, the threshold should be interpreted as a fixed methodological choice rather than an empirically optimal value [13, 14]. If a matrix contained fewer positive edges than required by the target density, all available positive edges were retained and the realized density was lower than the target.

In the resulting graph, nodes corresponded to EEG channels and edges corresponded to retained functional connectivity values. Edge weights represented connectivity strength. For metrics based on shortest paths, connectivity strength was converted into distance using the inverse of the edge weight, so that stronger functional connections corresponded to shorter graph distances.

2.4.1 Mean Connectivity

Mean connectivity was computed directly from the original, unthresholded connectivity matrix after symmetrization and removal of self-connections. It was defined as the average of all off-diagonal upper-triangular matrix entries:

$$\bar{A} = \frac{1}{N(N-1)/2} \sum_{i<j} A_{ij},$$

where N is the number of nodes and A_{ij} is the connectivity value between channels i and j.

This metric describes the overall level of functional coupling in a given network [4, 10]. Unlike the topology-based graph metrics, mean connectivity was not computed after thresholding. This was done to preserve information about the average connectivity strength across all channel pairs. For AEC matrices, negative correlations could therefore contribute to the mean connectivity value.

2.4.2 Density

Graph density describes the proportion of possible edges that are present in the thresholded graph [10, 12].. For an undirected graph with N nodes and E retained edges, density is defined as

$$D = \frac{E}{N(N-1)/2}.$$

In this thesis, a proportional threshold targeting a density of $20\,\%$ was applied before topology metrics were computed. Therefore, the density metric also served as a diagnostic measure confirming how many edges were retained after thresholding. In most cases, the realized density was expected to match the target density. If too few positive edges were available, the realized density could be lower.

2.4.3 Clustering Coefficient

The clustering coefficient measures the tendency of nodes to form locally interconnected groups [15, 16, 10]. In the context of EEG functional networks, a high clustering coefficient indicates that channels connected to the same channel also tend to be connected to each other, suggesting stronger local network organization.

For each thresholded weighted graph, the average weighted clustering coefficient was computed across all nodes:

$$C = \frac{1}{N} \sum_{i=1}^{N} C_i,$$

where C_i is the weighted clustering coefficient of node i. The use of weighted clustering allowed the metric to account not only for the presence of connections but also for their connectivity strength.

2.4.4 Global Efficiency

Global efficiency measures how efficiently information can be exchanged across the whole network [17, 10]. It is based on the inverse of the shortest-path distance between all pairs of nodes:

$$E_{\text{glob}} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}},$$

where d_{ij} is the shortest-path distance between nodes i and j.

In this thesis, shortest-path distances were computed using inverse connectivity strength as the edge distance. Therefore, stronger functional connections corresponded to shorter graph distances. If two nodes were disconnected, their contribution to global efficiency was set to zero. Higher global efficiency indicates that the network has shorter paths between nodes and may therefore support more integrated communication across the EEG channel network.

2.4.5 Characteristic Path Length

Characteristic path length measures the average shortest-path distance between nodes in a network [15, 10]. It provides a complementary view to global efficiency: whereas global efficiency increases when nodes are more easily reachable, characteristic path length decreases when the network becomes more integrated.

For a connected graph, characteristic path length is defined as

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij},$$

where d_{ij} is the shortest-path distance between nodes i and j.

Because thresholding may produce disconnected graphs, characteristic path length was computed on the largest connected component of each graph. This avoids undefined path lengths between disconnected node pairs. As with global efficiency, edge distances were defined as the inverse of connectivity strength.

2.4.6 Small-Worldness

Small-worldness quantifies whether a network combines high local clustering with short global path lengths compared with random reference networks [15, 18, 10]. This is relevant for brain network analysis because many functional brain networks have been described as balancing local specialization and global integration.

Small-worldness was computed as

$$\sigma = \frac{C/C_{\text{rand}}}{L/L_{\text{rand}}},$$

where C and L are the clustering coefficient and characteristic path length of the observed graph, and C_{rand} and L_{rand} are the corresponding mean values from randomized comparison graphs.

For each observed graph, random comparison graphs were generated by preserving the observed degree sequence through double-edge swaps when possible. The observed edge-weight distribution was then shuffled onto each randomized topology. In the final pipeline, 20 random graphs were used for each small-worldness estimate, with a fixed random seed to ensure reproducibility. Values greater than one indicate stronger small-world organization relative to the random comparison graphs.

2.4.7 Modularity

Modularity measures the extent to which a network can be divided into communities or modules [19, 10]. A module is a group of nodes that are more strongly connected to each other than to the rest of the network. In EEG functional networks, higher modularity may indicate stronger segregation of the network into relatively independent channel groups.

Modularity can be written conceptually as

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j),$$

where A_{ij} is the edge weight between nodes i and j, k_i and k_j are node strengths or degrees, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether nodes i and j belong to the same community.

In the implemented pipeline, modularity was computed using weighted community detection. When available, the Louvain community detection method was used; otherwise, a greedy modularity-based method was used as a fallback [20]. The same random seed was used to make the modularity calculation reproducible when stochastic community detection was applied.

2.4.8 Mean Betweenness Centrality

Betweenness centrality measures how often a node lies on shortest paths between other nodes [21, 10]. Nodes with high betweenness may act as bridges between different parts of the network. Betweenness centrality has also been used in EEG connectivity studies of drug-resistant epilepsy and neuromodulation [2]. In the present analysis, betweenness centrality was computed for each node and then averaged across all nodes to obtain a single network-level metric.

For a node v, betweenness centrality is defined as

$$BC(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths between nodes s and t, and $\sigma_{st}(v)$ is the number of those paths that pass through node v.

Shortest paths were computed using inverse connectivity strength as distance, so stronger edges corresponded to shorter paths. The mean betweenness centrality was then computed as

$$\overline{BC} = \frac{1}{N} \sum_{v=1}^{N} BC(v).$$

This produced a global summary of how strongly the network relied on central bridge-like nodes.

2.4.9 Assortativity

Assortativity measures the tendency of nodes to connect to other nodes with similar degree [22, 10]. A positive assortativity coefficient indicates that highly connected nodes tend to connect to other highly connected nodes, whereas a negative coefficient indicates that highly connected nodes tend to connect to less connected nodes.

Degree assortativity was computed for each thresholded graph. The coefficient can be interpreted as a correlation between the degrees of nodes at either end of an edge. Values close to one indicate assortative mixing, values close to zero indicate no clear degree-based mixing pattern, and values below zero indicate disassortative organization.

In the context of this thesis, assortativity was included as a global measure of network organization complementary to efficiency, clustering, and modularity. When assortativity was undefined, for example because the graph structure did not contain sufficient degree variability, the value was recorded as missing.

2.5 Output of the Connectivity and Graph Feature Pipeline

The connectivity and graph feature pipeline produced the network-level dataset used for statistical analysis. The final segmentation stage contained 632 saved temporal windows. Each window was analyzed across 6 frequency bands and 3 connectivity methods, giving

$$632 \times 6 \times 3 = 11\,376$$

expected connectivity matrices. No matrices failed validation, so the final connectivity output consisted of $11\,376$ valid 22×22 matrices.

Each matrix was then summarized by graph-theoretical features. The pipeline retained the subject/case identifier, seizure identifier, temporal window, frequency band, and connectivity method for every matrix, so each graph feature row remained traceable to its source EEG segment and connectivity definition.

The graph feature extraction stage produced one row per connectivity matrix, resulting in $11\,376$ graph-level observations. The extracted variables were mean connectivity, density, clustering coefficient, global efficiency, characteristic path length, small-worldness, modularity, mean betweenness centrality, and assortativity. Density was retained as a diagnostic measure, whereas the remaining graph metrics were used as outcome variables in the statistical analysis.

Additional diagnostic variables were stored for quality control and reproducibility. These included the threshold density, realized edge count, realized graph density, number of connected components, size of the largest connected component, number of random graphs used for small-worldness estimation, random seed, and software version information.

Thus, the final output of this stage was a seizure-window-level graph feature dataset. Each row represented one functional network defined by a specific subject/case identifier, seizure, temporal window, frequency band, and connectivity method. This dataset provided the input for the statistical analysis in the following chapter, where graph metrics were compared across Baseline, T0, T1, and T2.

Chapter 3

Statistical Analysis

3.1 Overview of the Statistical Design

The analysis was designed as an exploratory full-dataset statistical association analysis, to test whether network properties differed systematically between the interictal baseline window and the preictal windows T0, T1, and T2.

Each graph-level observation was derived from one functional connectivity network. A network was uniquely identified by the subject/case identifier, seizure identifier, temporal window, frequency band, connectivity method, and graph metric. The statistical analysis preserved this structure so that graph metrics were not treated as independent biological samples simply because they were produced across multiple frequency bands, connectivity methods, or network measures.

The main experimental factor was temporal window, with four levels: Baseline, T0, T1, and T2. Baseline represented an interictal reference window, whereas T0, T1, and T2 represented consecutive preictal windows approaching seizure onset. The statistical question was whether graph metrics varied across these windows, and in particular whether the preictal windows differed from Baseline.

The primary analysis was performed at the seizure level. In this analysis, each seizure was treated as the repeated-measures unit because the same seizure contributed graph metrics across multiple temporal windows. Seizures were nested within subject/case identifiers to account for the fact that multiple seizures could come from the same CHB-MIT case folder. This design helped avoid treating seizures from the same subject/case identifier as fully independent observations.

In addition to the seizure-level analysis, a subject-average sensitivity analysis was performed. In this analysis, graph metrics were first averaged across seizures within each subject/case identifier for each temporal window, frequency band, connectivity method, and graph metric. This produced one average trajectory per subject/case identifier. The purpose of this analysis was to evaluate whether effects observed at the seizure level were also visible when each subject/case identifier contributed equally to the group-level analysis.

For each graph metric, connectivity method, and frequency band, three types of statistical tests were performed. First, an omnibus window-effect test assessed whether there was evidence of any difference across Baseline, T0, T1, and T2. Second, planned post-hoc paired comparisons tested Baseline against each preictal window. Third, a preictal trend analysis examined whether graph metrics changed monotonically across T0, T1, and T2 as seizure onset approached.

Overall, the statistical design combined a primary seizure-level analysis with a subject-average sensitivity analysis. This allowed the thesis to examine both seizure-level temporal dynamics and the robustness of these effects at the subject/case level, while maintaining a clear distinction between exploratory statistical association and predictive modelling.

3.2 Data Organization for Statistical Testing

The statistical analysis used the graph feature outputs described in section 2.5. Each row of the original graph feature table represented one functional network derived from a specific combination of subject/case identifier, seizure, temporal window, frequency band, and connectivity method. The graph metrics were then converted into a long-format table, where each row contained one value for one graph metric.

The long-format organization made it possible to analyze each graph metric, connectivity method, and frequency band combination separately. For every such combination, the statistical model compared values across temporal windows. This avoided combining graph metrics with different meanings, connectivity methods with different physiological interpretations, or frequency bands with different spectral content.

The identifiers retained for each statistical row were: subject/case identifier, seizure identifier, temporal window, frequency band, connectivity method, graph metric. graph metric value.

This organization preserved the hierarchical structure of the dataset. Observations were grouped by seizure and subject/case identifier rather than treated as unrelated measurements.

3.2.1 Seizure-Level Analysis

The seizure-level analysis was the primary statistical analysis. In this analysis, each seizure contributed repeated measurements across temporal windows. For a given graph metric, connectivity method, and frequency band, the same seizure could contribute values for Baseline, T0, T1, and T2. This made it possible to test whether the graph metric changed within seizures as the EEG moved from the interictal baseline window toward seizure onset.

In the primary model, seizures were treated as repeated-measures units nested within subject/case identifiers. This was important because several CHB-MIT case folders contain multiple seizures. Treating all seizure-level rows as fully independent would overstate the amount of independent biological information in the dataset. The mixed-model structure was therefore used to account for repeated seizure observations while retaining the seizure-level temporal information.

The seizure-level analysis is useful because it preserves variability between seizures within the same subject/case identifier. This is relevant for epilepsy data, where different seizures from the same individual may show different preictal dynamics. However, because some subject/case identifiers contributed more seizures than others, a subject-average sensitivity analysis was also performed.

3.2.2 Subject-Average Sensitivity Analysis

The subject-average sensitivity analysis was performed to evaluate whether patterns observed at the seizure level were also visible when each subject/case identifier contributed equally. In this analysis, graph metric values were first averaged across seizures within each subject/case identifier for each temporal window, frequency band, connectivity method, and graph metric.

The resulting subject-average table contained one average value per subject/case identifier and temporal window for each metric–method–band combination. Standard deviation across seizures and the number of contributing seizures were also retained as descriptive information. The statistical tests were then repeated using these subject-level average trajectories.

This analysis reduced the influence of subject/case identifiers with many seizures. For example, a subject/case identifier contributing many usable seizures could strongly influence a seizure-level analysis, whereas in the subject-average analysis it contributed one average trajectory, like every other subject/case identifier. The subject-average approach therefore provided a complementary check on whether the main findings reflected consistent subject-level patterns rather than being driven mainly by subjects with larger numbers of seizures.

Averaging across seizures reduces within-subject seizure variability and may hide seizure-specific preictal effects. For this reason, the thesis reports the seizure-level analysis as the primary analysis and uses the subject-average analysis to assess robustness at the subject/case level.

3.3 Outcome Variables and Experimental Factors

The statistical analysis was organized around graph-level outcome variables and a fixed set of experimental factors. Each outcome variable represented one graph-theoretical feature extracted from a functional connectivity network. The experimental factors described how each network was generated: the temporal window, frequency band, and connectivity method.

The outcome variables analyzed in this thesis were the graph metrics described in section 2.4.

Although graph density was computed and retained as a diagnostic variable, it was not treated as a primary statistical outcome. This is because the graph construction procedure applied a proportional threshold targeting a fixed density. Therefore, density mainly served as a quality-control measure confirming the realized number of retained edges after thresholding.

The statistical tests therefore evaluated whether graph metrics changed from Baseline to the preictal period and whether there was evidence of progressive change within the preictal interval.

The second experimental dimension was frequency band. The analysis was performed separately for the six frequency bands used throughout the pipeline: delta, theta, alpha1, alpha2, beta1, and beta2. This band structure follows the pre-seizure graph-theoretical analysis of Vecchio et al., who analyzed connectivity separately across the same frequency ranges [3]. The bands were not pooled in the statistical models because seizure-related network organization may differ across frequency ranges, and combining bands could obscure frequency-specific effects [23, 2].

The third experimental dimension was connectivity method. The analysis was performed separately for coherence, wPLI, and AEC networks. Separate statistical testing by method allowed to evaluate whether observed window effects were specific to a particular definition of functional connectivity.

For each combination of graph metric, connectivity method, and frequency band, a separate statistical analysis was performed. Since the final analysis included 8 graph metrics, 3 connectivity methods, and 6 frequency bands, this produced 144 metric–method–band combinations. Each combination was analyzed independently across temporal windows. This design made it possible to identify which graph properties, connectivity definitions, and frequency bands showed evidence of preictal change.

The statistical models did not treat individual epochs, channels, graph metrics, frequency bands, or connectivity methods as independent biological samples. Instead, the unit of repeated measurement was defined at the seizure level for the primary analysis and at the subject/case level for the sensitivity analysis. This distinction was important because the large number of derived graph features does not imply an equally large number of independent subjects or seizures.

3.4 Omnibus Testing Across Temporal Windows

The first statistical question was whether each graph metric showed an overall effect of temporal window. This was tested using an omnibus window-effect test for each graph metric, connectivity method, and frequency band combination.

The purpose of the omnibus test was to determine whether there was evidence of any difference across the four temporal windows, without first specifying which preictal window differed from Baseline. This step was important because the analysis involved many graph metrics, connectivity methods, and frequency bands. Planned pairwise comparisons were therefore interpreted mainly when the corresponding omnibus test indicated a significant window effect after multiple-comparison correction.

For the primary seizure-level analysis, the omnibus test was performed using a linear mixed-effects model. This modelling framework was selected because it is appropriate for repeated-measures and grouped data, allowing temporal-window effects to be estimated while accounting for the non-independence of seizures within subject/case identifiers [24, 25]. For each metric–method–band combination, temporal window was included as a categorical fixed effect:

$$y_{s,w} = \beta_0 + \beta_w + u_{\text{subject}} + u_{\text{seizure}} + \varepsilon_{s,w},$$

where $y_{s,w}$ is the graph metric value for seizure s in window w, β_0 is the intercept, β_w represents the fixed effect of temporal window, u_{subject} accounts for grouping by subject/case identifier, u_{seizure} accounts for repeated measurements within seizure, and $\varepsilon_{s,w}$ is the residual error term.

Baseline was used as the reference window in the mixed model. The omnibus window effect was then evaluated using a Wald chi-square test of the window terms [26]. This tested whether the temporal window coefficients were jointly different from zero. A significant result indicated that the graph metric varied across Baseline, T0, T1, and T2 for the corresponding connectivity method and frequency band.

When the mixed-effects model could not be estimated, the pipeline attempted a repeated-measures fallback analysis. If the Pingouin statistical package was available, a one-way repeated-measures ANOVA was attempted; otherwise, a manual one-way repeated-measures ANOVA was available as a secondary fallback [27, 28]. This fallback preserved the within-unit repeated-measures structure, although it did not explicitly model the nesting of seizures within subject/case identifiers. Cases in which neither the mixed-effects model nor the fallback analysis produced a valid test statistic were marked as non-estimable and excluded from inferential interpretation.

The same omnibus testing logic was also applied to the subject-average sensitivity analysis. In that case, the repeated-measures unit was the subject/case identifier rather than the seizure. Each subject/case identifier contributed one average value per window for a given metric–method–band combination. The subject-average omnibus model therefore tested whether the average subject-level trajectory differed across Baseline, T0, T1, and T2.

For each omnibus test, the pipeline recorded the number of complete repeated observations, the number of contributing subject/case identifiers, the model type, the test statistic, degrees of freedom where applicable, the uncorrected p-value, the FDR-corrected p-value, and any fallback or skip reason. Tests were skipped when the number of complete repeated observations was below the predefined minimum required for analysis.

3.5 Post-hoc Baseline-vs-Preictal Comparisons

After the omnibus temporal-window tests, planned post-hoc comparisons were performed to identify which preictal windows differed from the interictal baseline condition. These comparisons were defined a priori because the central research question of the thesis concerned whether graph-theoretical network features changed before seizure onset relative to Baseline.

For each graph metric, connectivity method, and frequency band combination, three paired Baseline-vs-preictal contrasts were tested: Baseline vs. T0, Baseline vs. T1, Baseline vs. T2.

These contrasts compared the interictal reference window with each of the three consecutive preictal windows. T0 represented the earliest preictal window, T1 the middle preictal window, and T2 the window closest to seizure onset. This design allowed the analysis to evaluate whether a metric differed from Baseline at any specific preictal interval.

For each paired contrast, the primary post-hoc test was a paired-samples t-test. If $x_{\text{Baseline},s}$ denotes the baseline value for seizure s and $x_{\text{Preictal},s}$ denotes the corresponding preictal value, the paired difference was defined as

$$d_s = x_{\text{Preictal},s} - x_{\text{Baseline},s}.$$

The paired t-test evaluated whether the mean of these within-seizure differences differed from zero. A positive mean difference indicated that the metric was higher in the preictal window than in Baseline, whereas a negative mean difference indicated that the metric was lower in the preictal window.

For each contrast, the analysis recorded the number of paired observations, the number of contributing subject/case identifiers, the Baseline mean, the comparison-window mean, the mean difference, the t statistic, degrees of freedom, the uncorrected p-value, and the FDR-corrected p-value. Cohen's d_z was also computed as an effect-size measure for paired data [29]:

$$d_z = \frac{\bar{d}}{s_d},$$

where $\bar{d}$ is the mean paired difference and s_d is the standard deviation of the paired differences.

As a non-parametric sensitivity check, a Wilcoxon signed-rank test was also computed for each paired contrast when possible [30]. The Wilcoxon signed-rank test does not require the paired differences to be normally distributed, although its standard interpretation assumes that the distribution of paired differences is approximately symmetric. Therefore, it provided an additional check on whether the paired comparison was robust to the normality assumption of the paired t-test. Both the paired t-test and Wilcoxon test results were retained in the statistical output. Both the paired t-test and Wilcoxon test results were retained in the statistical output.

The same planned post-hoc contrasts were repeated for the subject-average sensitivity analysis. In that analysis, the paired observations were subject/case-level average values rather than individual seizure-level values. Thus, each subject/case identifier contributed at most one paired value to each Baseline-vs-preictal contrast.

Post-hoc comparisons were interpreted cautiously because a large number of contrasts were tested across graph metrics, connectivity methods, and frequency bands. In the final interpretation, post-hoc results were considered most meaningful when the corresponding omnibus temporal-window effect was significant after Benjamini–Hochberg FDR correction. This rule reduced the risk of over-interpreting isolated pairwise differences that were not supported by an overall window effect.

3.6 Preictal Trend Analysis

In addition to comparing each preictal window with Baseline, the statistical pipeline tested whether graph metrics changed progressively during the preictal period itself. This analysis used only the three preictal windows T0, T1, and T2. Baseline was not included because the purpose of this test was to evaluate directional change as seizure onset approached, rather than to compare the preictal period with the interictal reference condition.

The three preictal windows were represented by their midpoint times relative to seizure onset. T0 was assigned a time of -7.5 minutes, T1 a time of -4.5 minutes, and T2 a time of -1.5 minutes. These values correspond to the centers of the three consecutive three-minute preictal windows. A positive slope therefore indicates that the graph metric increased as the recording approached seizure onset, whereas a negative slope indicates that the metric decreased across the preictal period.

For each graph metric, connectivity method, and frequency band combination, the primary preictal trend analysis used a linear mixed-effects model when available. The model can be written as

$$y_{s,t} = \beta_0 + \beta_1 t + u_{\text{subject}} + u_{\text{seizure}} + \varepsilon_{s,t},$$

where $y_{s,t}$ is the graph metric value for seizure s at preictal time t, β_0 is the intercept, β_1 is the fixed slope across preictal time, u_{subject} accounts for subject/case-level grouping, u_{seizure} accounts for repeated measurements within seizure, and $\varepsilon_{s,t}$ is the residual error term.

The main parameter of interest was the slope coefficient β_1. This coefficient describes the average change in the graph metric per minute during the preictal interval. Because time was coded in minutes relative to seizure onset, the sign of the slope must be interpreted with the direction of time in mind. A positive slope means that the metric increased from T0 toward T2, while a negative slope means that the metric decreased as seizure onset approached.

Only seizures with complete T0, T1, and T2 values were included in the seizure-level preictal trend analysis. This ensured that the trend was estimated from within-seizure trajectories across all three preictal windows. The same complete-case logic was applied to each metric–method–band combination separately.

When the mixed-effects model could not be used, the pipeline applied a fallback trend analysis. In the fallback approach, a linear slope was first estimated separately for each seizure across T0, T1, and T2. These seizure-level slopes were then tested against zero using a one-sample t-test. This preserved the interpretation of the trend as a within-seizure temporal change, although it did not explicitly model subject/case-level grouping.

The trend analysis was intended to complement the omnibus and post-hoc tests. The omnibus test assessed whether any difference existed across Baseline, T0, T1, and T2. The post-hoc tests identified which preictal windows differed from Baseline. The trend test asked a different question: whether the metric changed systematically within the preictal period itself. Therefore, a metric could show a Baseline-vs-preictal difference without a monotonic T0–T2 trend, or it could show a preictal trend without a strong Baseline contrast.

For each trend test, the statistical output recorded the number of complete preictal seizures, the number of contributing subject/case identifiers, the model type, the estimated slope per minute, the test statistic, the uncorrected p-value, the FDR-corrected p-value, and any fallback or skip reason. As with the other statistical outputs, trend results were interpreted after correction for multiple comparisons.

3.7 Multiple-Comparison Correction

Because graph metrics were tested separately across connectivity methods and frequency bands, the analysis involved many statistical tests. To reduce the risk of false-positive interpretation, Benjamini–Hochberg false discovery rate (FDR) correction was applied [31]. FDR correction controls the expected proportion of false discoveries among results declared significant, which is appropriate for an exploratory analysis with many related tests.

Correction was applied separately within each output family: seizure-level omnibus tests, seizure-level post-hoc tests, seizure-level preictal trend tests, subject-average omnibus tests, and subject-average post-hoc tests. Thus, correction was performed across tests belonging to the same statistical family, rather than separately for each subject/case identifier or each individual graph metric.

For post-hoc analyses, paired t-test p-values and Wilcoxon signed-rank p-values were corrected separately. A corrected significance threshold of $\alpha = 0.05$ was used. Uncorrected p-values were retained for transparency, but interpretation focused on FDR-corrected values. Planned post-hoc comparisons were interpreted primarily when the corresponding omnibus window-effect test also survived FDR correction.

3.8 Validation and Output of the Statistical Pipeline

The statistical pipeline transformed the graph feature table into a long-format dataset suitable for repeated-measures testing. The input contained $11\,376$ graph-level rows and 8 analyzed graph metrics, resulting in $91\,008$ long-format observations. Each observation retained its subject/case identifier, seizure identifier, temporal window, frequency band, connectivity method, graph metric, and value.

The final dataset included 24 subject/case identifiers and 159 usable seizures. Of these, 155 seizures had complete values across Baseline, T0, T1, and T2 and were used for Baseline-inclusive seizure-level omnibus testing. All 159 seizures had complete T0, T1, and T2 values and were available for the preictal trend analysis.

The analysis covered 144 metric–method–band combinations (8 graph metrics $\times$ 3 connectivity methods $\times$ 6 frequency bands). At the seizure level, this produced 144 omnibus tests, 432 planned Baseline-vs-preictal post-hoc contrasts, and 144 preictal trend tests. The subject-average sensitivity analysis produced $13\,824$ rows, 112 omnibus tests with valid model output, and 432 post-hoc contrasts.

The output tables included descriptive summaries, seizure-level omnibus tests, seizure-level post-hoc comparisons, seizure-level preictal trends, subject-average summaries, subject-average omnibus tests, and subject-average post-hoc comparisons. Inferential outputs retained the model type, test statistic, uncorrected and FDR-corrected p-values, effect-size information where applicable, and diagnostic fields such as fallback or skip reasons.

These validation outputs confirmed that the dataset structure was preserved from graph extraction to statistical testing and provided the basis for the results chapter.

Chapter 4

Experimental Results

4.1 Overview of the Statistical Output

This chapter reports the results of the statistical analysis applied to the graph-theoretical EEG connectivity features. The aim was to evaluate whether network properties differed across the temporal windows Baseline, T0, T1, and T2, and whether any observed effects were consistent across seizure-level and subject-average analyses.

The final statistical input contained $11\,376$ graph-level observations. Eight graph metrics were analyzed, producing $91\,008$ long-format rows. The dataset included 24 subject/case identifiers and 159 usable seizures. Of these, 155 seizures had complete values across all four windows and were included in Baseline-inclusive seizure-level omnibus testing. All 159 seizures had complete values across the three preictal windows T0, T1, and T2 and were therefore available for the preictal trend analysis.

The analysis was performed separately for each combination of graph metric, connectivity method, and frequency band. Since 8 graph metrics, 3 connectivity methods, and 6 frequency bands were analyzed, the full seizure-level analysis included 144 metric–method–band combinations. This produced 144 omnibus window-effect tests, 432 planned Baseline-vs-preictal post-hoc contrasts, and 144 preictal trend tests.

The subject-average sensitivity analysis produced 13 824 subject-average rows. In this analysis, each subject/case identifier contributed one average trajectory for each metric, method, band, and window combination. The subject-average analysis produced 112 omnibus tests with valid model output and 432 planned post-hoc contrasts.

After Benjamini–Hochberg FDR correction, the main finding was a temporal-window effect for global efficiency computed from delta-band coherence networks. This effect was present in both the seizure-level omnibus analysis and the subject-average omnibus analysis. At the seizure level, the corresponding post-hoc analysis localized the strongest difference to the Baseline-vs-T1 contrast. No preictal trend test survived FDR correction, and no subject-average post-hoc contrast survived FDR correction.

Analysis family	FDR-significant result
Seizure-level omnibus tests	Global efficiency, coherence, delta band
Seizure-level post-hoc contrasts	Global efficiency, coherence, delta band, Baseline vs T1
Seizure-level preictal trend tests	No FDR-significant trend effects
Subject-average omnibus tests	Global efficiency, coherence, delta band
Subject-average post-hoc contrasts	No FDR-significant post-hoc contrasts

Table 4.1: Summary of FDR-significant findings across the main statistical output families.

Overall, the results indicate that the most robust statistical effect in the full dataset was an increase in delta-band coherence global efficiency during the preictal period, especially in the T1 window. The following sections report the seizure-level omnibus tests, post-hoc contrasts, preictal trend results, and subject-average sensitivity analysis in more detail.

4.2 Seizure-Level Omnibus Window Effects

The seizure-level omnibus analysis tested whether each graph metric showed an overall difference across Baseline, T0, T1, and T2. A separate test was performed for each metric–method–band combination, resulting in 144 omnibus window-effect tests.

After Benjamini–Hochberg FDR correction, one seizure-level omnibus effect remained statistically significant. This effect was observed for global efficiency computed from delta-band coherence networks. The test included 155 complete seizures from 24 subject/case identifiers and showed a significant effect of temporal window ($\chi^2(3) = 21.11$, $p_{\text{uncorrected}} = 0.000100$, $p_{\text{FDR}} = 0.014367$).

Metric	Method	Band	Test	n	χ^2	p	p_{FDR}
Global efficiency	Coherence	Delta	Window effect	155	21.11	0.000100	0.014367

Table 4.2: FDR-significant seizure-level omnibus window effect.

The descriptive means for this metric showed an increase from Baseline to the preictal windows. Mean delta-band coherence global efficiency was $0.224\,424$ at Baseline, $0.228\,493$ at T0, $0.234\,382$ at T1, and $0.233\,047$ at T2. The highest mean value was therefore observed in T1, followed by a small decrease in T2.

Window	n	Mean	SD	95% CI
Baseline	155	0.224424	0.036161	[0.218687, 0.230162]
T0	159	0.228493	0.036656	[0.222751, 0.234234]
T1	159	0.234382	0.033300	[0.229166, 0.239598]
T2	159	0.233047	0.033635	[0.227779, 0.238316]

Table 4.3: Descriptive statistics for seizure-level delta-band coherence global efficiency across temporal windows.

No other seizure-level omnibus test survived FDR correction. Therefore, the only robust seizure-level evidence for a temporal-window effect was found for global efficiency in delta-band coherence networks. The corresponding planned post-hoc comparisons are reported in the next section.

4.3 Post-hoc Baseline-vs-Preictal Comparisons

Planned post-hoc comparisons were used to determine which preictal windows differed from Baseline. These comparisons were interpreted primarily for metric–method–band combinations that showed a significant omnibus window-effect after FDR correction.

Only one post-hoc contrast survived FDR correction. This was the Baseline-vs-T1 comparison for global efficiency in delta-band coherence networks. The paired comparison included 155 seizures from 24 subject/case identifiers. Mean global efficiency increased from $0.224\,424$ at Baseline to $0.233\,575$ at T1, corresponding to a mean paired difference of $0.009\,150$. This contrast was significant after FDR correction ($t(154) = 4.22$, $p_{\text{uncorrected}} = 0.000042$, $p_{\text{FDR}} = 0.018111$, Cohen's $d_z = 0.339$).

Contrast	Baseline mean	Preictal mean	Difference	t	p	p_{FDR}
Baseline vs T0	0.224424	0.227550	0.003126	1.35	0.177660	0.600708
Baseline vs T1	0.224424	0.233575	0.009150	4.22	0.000042	0.018111
Baseline vs T2	0.224424	0.231975	0.007551	3.08	0.002427	0.288818

Table 4.4: Seizure-level post-hoc comparisons for delta-band coherence global efficiency.

The Baseline-vs-T0 comparison was not significant, indicating that the earliest preictal window did not differ clearly from Baseline for this metric. The Baseline-vs-T2 comparison showed an uncorrected difference, but it did not survive FDR correction. Therefore, the only statistically robust localized post-hoc effect was the increase from Baseline to T1.

The Wilcoxon signed-rank sensitivity analysis showed the same direction of change for the Baseline-vs-T1 contrast, but its FDR-corrected value was just above the significance threshold ($p_{\text{uncorrected}} = 0.000121$, $p_{\text{FDR}} = 0.052186$). Thus, the post-hoc evidence was strongest in the paired t-test, while the non-parametric sensitivity result supported the same direction but did not meet the corrected threshold.

No other Baseline-vs-preictal contrast survived FDR correction. Therefore, the post-hoc analysis localized the main seizure-level omnibus effect to a Baseline-to-T1 increase in delta-band coherence global efficiency.

4.4 Preictal Trend Analysis

The preictal trend analysis tested whether graph metrics changed linearly across T0, T1, and T2 as seizure onset approached. Unlike the Baseline-vs-preictal comparisons, this analysis did not include the Baseline window. It focused only on directional change within the preictal period.

A total of 144 seizure-level trend tests were performed, one for each metric–method–band combination. All 159 usable seizures contributed to the trend analysis because complete T0, T1, and T2 values were available for all seizures.

No preictal trend survived Benjamini–Hochberg FDR correction. Therefore, the analysis did not provide corrected evidence for a consistent monotonic increase or decrease across T0, T1, and T2 for any graph metric, connectivity method, or frequency band.

For the main omnibus finding, delta-band coherence global efficiency showed a positive preictal slope, indicating higher values closer to seizure onset. However, this trend did not survive FDR correction ($\beta = 0.000759$ per minute, $p_{\text{uncorrected}} = 0.030552$, $p_{\text{FDR}} = 0.347761$).

Metric	Method	Band	n	Slope/min	p	p_{FDR}
Global efficiency	Coherence	Delta	159	0.000759	0.030552	0.347761

Table 4.5: Preictal trend result for the main significant omnibus metric.

These results suggest that the significant temporal-window effect observed for delta-band coherence global efficiency was not explained by a strictly monotonic T0–T2 trajectory. Instead, the main seizure-level effect was better localized by the Baseline-vs-preictal post-hoc analysis, especially the Baseline-vs-T1 comparison.

4.5 Subject-Average Sensitivity Analysis

The subject-average sensitivity analysis tested whether the main seizure-level finding was also visible when each subject/case identifier contributed one average trajectory. This analysis reduces the influence of subjects with many seizures and provides a complementary subject-level view of the results.

The subject-average omnibus analysis identified one FDR-significant temporal-window effect. As in the seizure-level analysis, this effect was found for global efficiency computed from delta-band coherence networks. The model included all 24 subject/case identifiers and showed a significant effect of temporal window ($\chi^2(3) = 24.04$, $p_{\text{uncorrected}} = 0.000025$, $p_{\text{FDR}} = 0.002748$).

Metric	Method	Band	n	χ^2	p	p_{FDR}
Global efficiency	Coherence	Delta	24	24.04	0.000025	0.002748

Table 4.6: FDR-significant subject-average omnibus window effect.

The subject-average means followed the same general pattern as the seizure-level analysis. Mean global efficiency was $0.222\,440$ at Baseline, $0.223\,423$ at T0, $0.233\,735$ at T1, and $0.229\,908$ at T2. The highest average value was again observed in T1.

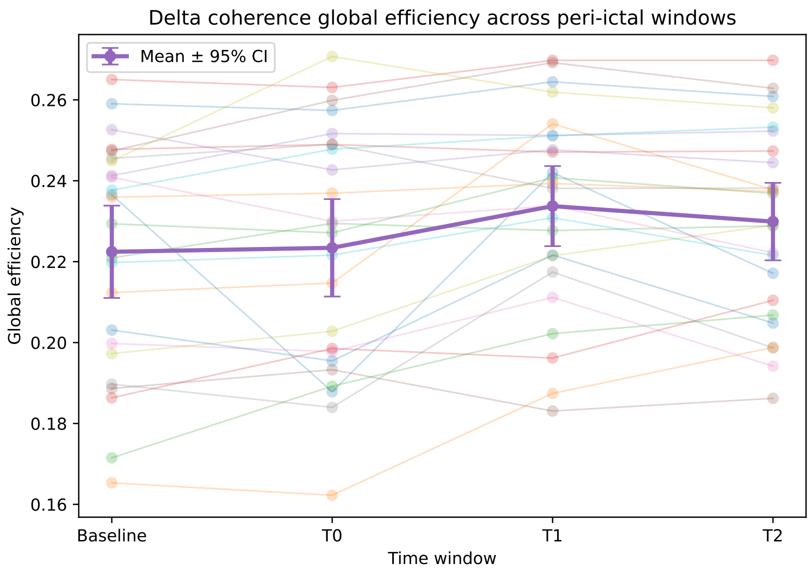

Figure 4.1: Subject-average trajectory of delta-band coherence global efficiency across Baseline, T0, T1, and T2. Each subject/case identifier contributes one averaged value per temporal window.

As shown in figure 4.1, the subject-average trajectory increased from Baseline toward the preictal windows, with the highest mean value in T1. This supports the seizure-level omnibus finding by showing that the same pattern was present after averaging across seizures within each subject/case identifier.

However, none of the subject-average Baseline-vs-preictal post-hoc contrasts survived FDR correction. The Baseline-vs-T1 contrast showed the strongest uncorrected effect ($t(23) = 4.40$, $p_{\text{uncorrected}} = 0.000206$, $p_{\text{FDR}} = 0.089102$, Cohen's $d_z = 0.899$), but it did not meet the corrected significance threshold. Therefore, the subject-average analysis confirmed the overall temporal-window effect but did not localize a specific Baseline-vs-preictal contrast after correction.

No other subject-average omnibus test survived FDR correction. Thus, the subject-average analysis supported the main result observed at the seizure level: delta-band coherence global efficiency was the only graph feature that showed a robust temporal-window effect across both analysis levels.

4.6 Summary of Main Findings

The full-dataset statistical analysis identified one robust graph-theoretical effect after FDR correction. Global efficiency computed from delta-band coherence networks showed a significant temporal-window effect in both the seizure-level and subject-average omnibus analyses.

At the seizure level, the effect was localized by the planned post-hoc comparisons to an increase from Baseline to T1. Mean global efficiency increased from $0.224\,424$ at Baseline to $0.233\,575$ at T1, and this contrast remained significant after FDR correction. The Baseline-vs-T2 comparison showed the same direction of change but did not survive correction.

The preictal trend analysis did not identify any FDR-significant monotonic T0–T1–T2 changes. Therefore, the main result should not be interpreted as a strict linear increase toward seizure onset. Instead, the strongest evidence was for a window-specific increase in delta-band coherence global efficiency, especially around the T1 interval.

The subject-average sensitivity analysis supported the main finding at the level of subject/case identifiers. The same metric–method–band combination showed a significant omnibus window effect after averaging seizures within each subject/case identifier. However, subject-average post-hoc contrasts did not survive FDR correction, indicating that the localized Baseline-vs-T1 effect was stronger at the seizure level than at the subject-average level.

No other graph metric, connectivity method, or frequency band combination survived FDR correction in the omnibus, post-hoc, or trend analyses. Overall, the results suggest that the most consistent preictal network alteration in this dataset was a change in global integration of delta-band coherence networks, rather than a broad effect across all graph metrics or all frequency bands.

Appendix A

Reproducibility

This appendix will document the computational environment, repository structure, pipeline scripts, parameter choices, and data-management decisions used in the thesis.

Raw EEG data, processed arrays, connectivity matrices, graph metrics, and large generated outputs are intentionally excluded from this writing repository.

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